Post-Quantum Authority Infrastructure for Agentic AI, IoT & Physical AI

Autonomous agents are gaining the ability to act, spend, and trigger downstream consequences across enterprise systems before the control layer around them has matured. Independent research and national cyber guidance now point to the same conclusion: existing identity, governance, and security frameworks were not built for autonomous agents and do not provide a durable solution for delegated authority, bounded execution, and accountable action. This document evaluates whether SHOGI's architecture solves that problem in a way existing approaches structurally cannot, and whether that distinction is strong enough to matter under enterprise and acquisition scrutiny. References to SHOGI's behavior describe the designed and patent-pending architecture, not a production deployment.

1. Mission Statement

"SHOGI is post-quantum cryptographic authority infrastructure for Agentic AI, IoT, and Physical AI. Every governed action is bound to a cryptographic identity and adjudicated against governance before execution. SHOGI establishes which entities hold authority, assembles consensus across agents, humans, or both, and permits execution only through an authorized, constrained path for every governed operation."

This mission statement is a technical commitment, not a positioning claim. Each of its four components is independently verifiable.

Cryptographic authority infrastructure. SHOGI will use post-quantum, Learning With Errors (LWE)-based Identity-Based Encryption as its cryptographic substrate. LWE is a foundational hardness assumption in modern lattice-based post-quantum cryptography, including the family of techniques reflected in NIST FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA). The claim is technically supportable.

Every governed action is identity-bound. SHOGI derives private keys from ordered identity strings that encode role, mission scope, authority tier, jurisdiction, and operational boundaries. Every governed action must stay within these limits. Critical actions also need a one-time permit, approved by a quorum and verified before execution.

An agent with ActionScope=ReadPropose cannot independently supply valid cryptographic authority for credential harvesting, lateral movement, or any action beyond its defined scope. The identity dictates the policy, not a rule applied after key generation. The claim is architecturally coherent.

Governance-enforced before execution. SHOGI's pre-execution authorization layer adjudicates authority after action formation but before any state commitment. The enforcement layer does not participate in execution logic and cannot be used as an execution primitive by an attacker. This distinction materially changes the security guarantee.

Scope of the guarantee. This guarantee applies to every governed operation within SHOGI's control boundary, provided the key authority remains trusted and quorum rules are configured correctly. The primary operational failure mode is misconfigured quorum seats, which can make evidence requirements bypassable by design; Section 4 addresses that risk.

2. Core Problem Definition

Multiple independent research papers establish the problem SHOGI is built to solve. Agents of Chaos (arXiv, Feb 2026) red-teamed autonomous AI agents in live environments and documented unauthorized compliance with non-

owners, identity spoofing, cross-agent propagation of unsafe behavior, and partial system takeover. The authors concluded that accountability, delegated authority, and responsibility for downstream harms remain entirely unresolved. AI Identity: Standards, Gaps, and Research Directions for AI Agents (arXiv, Apr 2026) then evaluated every major AI identity framework and found that none were built for autonomous agents, and none solve the identity and authentication challenge. These findings point to an architectural gap that requires more than incremental improvements to existing controls. A third independent paper, Agent Security is a Systems Problem (arXiv, May 2026) reaches the same conclusion from a systems-security perspective: robust agent security requires hard enforcement of least privilege, complete mediation, and instruction/data separation at the system boundary, not soft controls inside the model context. As national cyber agencies now recognize, agentic AI is advancing faster than the governance and control frameworks required to deploy it safely. In May 2026, cybersecurity agencies from the United States, United Kingdom, Australia, Canada, and New Zealand jointly issued Careful adoption of agentic AI services, warning that agentic AI is already entering critical infrastructure and defense environments and can introduce service disruption, privacy breaches, and cybersecurity incidents. The guidance states that organizations should never grant agentic AI broad or unrestricted access to sensitive data or critical systems, and should confine these systems to low-risk, non-sensitive tasks. It also recommends human review or approval before high-impact actions. Its emphasis on identity management, least privilege, oversight, containment, and auditability supports SHOGI's thesis that autonomous agents require a cryptographic trust and control layer, not just model intelligence.

Emerging Zero Trust frameworks for AI agents reinforce the same directional conclusion as the research and national guidance cited above: autonomous systems require verifiable identity, least privilege, scoped access, continuous authorization, and stronger controls over high-risk actions. SHOGI aligns with that diagnosis but proposes a distinct enforcement mechanism. Rather than relying solely on software policy applied to authenticated sessions or requests, SHOGI is designed to bind governance constraints to Ordered Identities and cryptographic key material, with authority adjudicated before execution.

The OWASP Top 10 for Agentic Applications for 2026 reaches a similar conclusion from an application-security perspective. OWASP identifies Agent Goal Hijack, Identity and Credential Compromise, Tool Misuse, Agentic Supply Chain Vulnerabilities, and multi-agent failure propagation as among the highest-priority risks facing autonomous systems. These categories reinforce the same underlying observation: autonomous agents require verifiable authority boundaries that remain enforceable even when models, prompts, memory, tools, or external dependencies are compromised. SHOGI is designed around that assumption. Rather than attempting to make agent reasoning trustworthy, SHOGI constrains which actions can be validly authorized, regardless of how the reasoning process arrived at them.

The market consequence of leaving this unsolved is now visible. In late April 2026, Experian launched Agent Trust with Visa, Cloudflare, and Skyfire, signaling that major identity, payments, and network players now view agent verification as foundational infrastructure for agentic commerce. Digital Commerce 360, citing Gartner's IT Symposium/Xpo 2025 forecasts, reported that AI agents could intermediate more than $15 trillion in B2B spending by 2028. Agent Trust confirms that agent identity is becoming foundational infrastructure. It also confirms exactly where the hard problem begins: proving what an agent is authorized to do, under whose authority, and within what enforceable limits.

Agent Trust tells you a verified human authorized the agent. SHOGI is designed for the layer that comes next. With SHOGI, an agent cannot independently supply valid cryptographic authority for an action outside its provisioned scope. The restriction is bound to its Ordered Identity and key material, not merely expressed as a software policy rule. SHOGI's use of post-quantum cryptography is intended to preserve this authority boundary against both classical and quantum cryptanalytic threats.

Why control-plane / user-plane separation fails in agentic systems

In traditional software, control-plane instructions and user-plane data flow through structurally separate channels. An SQL parser distinguishes keywords from values through grammar. The boundary is architectural.

In LLM-based agentic systems, both channels flow through the same pathway: the context window. As NVIDIA's research confirms, "both flavors of prompt injection are rooted in the lack of separation between the control and data plane in the LLM architecture." This is not a bug. It is the design. The model's ability to follow diverse natural-language instructions is inseparable from its vulnerability to adversarial instructions embedded in data.

Why prompt-level governance is non-binding

System prompts, instruction hierarchies, and RLHF-based alignment apply constraints within the model's context. Independent research found 73% of tested agentic platforms failed to enforce at least one boundary. OpenAI acknowledged in December 2025 that prompt injection "is unlikely to ever be fully solved" because trusted and untrusted inputs share the same context window.

These controls operate on the attack surface, not outside it. A sufficiently crafted adversarial input can cause any soft constraint to yield. This is not a question of defense depth. It is a question of where the enforcement point lives.

Why authentication alone does not preserve action authority

Authenticating an agent does not establish that every subsequent action is authorized. A compromised policy engine, a prompt-injected agent presenting a valid identity, or a misconfigured rule set can permit an action outside its intended scope. SHOGI addresses this risk by binding action authority to Ordered Identities and requiring cryptographically valid authorization before a governed action can execute.

Why token-based approaches do not preserve authority integrity

Bearer tokens operate on one principle: possession equals authorization. In agentic systems operating at machine speed, tokens traverse distributed environments outside central security visibility. Replay attack success rates have reached 66% in controlled agentic environments.

Proof-of-Possession tokens (DPoP, RFC 9449) improve on this by binding a token to the cryptographic key held by the agent. Two structural limitations remain:

• PoP binds a token to a cryptographic key, but that binding alone does not establish whether a specific action satisfies the required governance constraints.

• Token-based systems can support fine-grained authorization. SHOGI’s proposed distinction is binding authority constraints to Ordered Identities and key material, then requiring action-specific authorization through its protected execution path.

Is this a load-bearing architectural flaw?

Yes. The inability to separate control-plane instructions from user-plane data in LLM-based systems is not a surface vulnerability. It is a consequence of how these models process information. Any governance approach operating inside the context window inherits this vulnerability. This is the independent finding of NVIDIA, OWASP, and academic security literature, not SHOGI's framing of the problem.

Moving the enforcement point outside the context window to a pre-execution, cryptographically bound layer is a structurally appropriate response to a structurally defined problem.

3. The SHOGI System:

SHOGI's cryptographic foundation is LWE-based Identity-Based Encryption. In standard PKI, a certificate authority issues a certificate binding a key pair to an identity. In IBE, the identity itself functions as the public key, and anyone with the system's public parameters can encrypt directly to that identity without exchanging certificates. The Private Key Generator (PKG) holds a master secret and derives per-identity private keys.

SHOGI extends this model with ordered identities: deterministic strings assembled from characteristic values such as role, mission scope, authority tier, jurisdiction, and operational state, using an ordering function that produces a consistent representation regardless of input sequence. This lets governance constraints be embedded directly into the key material rather than attached later as policy metadata. As a result, SHOGI's guardrails are cryptographic properties of the keys themselves, not references to an external software policy engine, which makes them materially harder to bypass or tamper with.

The ordered identity mechanism is patent-pending, described in U.S. Patent Application No. 19/337,564 (METHOD AND SYSTEM FOR SECURE DATA COMMUNICATION USING ORDERED IDENTITIES) and in PCT Application No. PCT/US2025/047587. In the PCT application, the USPTO, acting as International Searching Authority, issued a favorable written opinion assessing all 25 claims as meeting the criteria for novelty, inventive step, and industrial applicability, with no negative assessment on any claim. The examiner considered Gentry (US 2008/0052521), IBM (US 2019/0173666), and Cisco (US 2023/0308439), classified all three as general background art (Category "A"), and concluded that they do not, alone or combined, disclose or render obvious the claimed combination. The reasoning centers on ordering identity characteristics into a consistent concatenated string and on validation based on those ordered identities.

Lead IP counsel Marc Kaufman, who has 30 years of patent experience, considers this a rare outcome, believes the same examiner is likely to handle the U.S. application, and, based on prior art searches and this opinion, expects the ordered identity mechanism to be granted. The opinion is preliminary and non-binding, and formal examination of the U.S. and national-phase applications will proceed separately.

Non-IBE schemes, whether classical or post-quantum, do not make guardrails cryptographic properties of the keys themselves. Their guardrails are references to an external software policy engine.

Identity Is the Policy

An agent with ActionScope=ReadPropose receives a private key bound to that ordered identity, not to an identity authorizing write or delete operations. SHOGI's protected execution path is designed to require cryptographic authority corresponding to the requested action. The agent may formulate an out-of-scope request, but it cannot independently supply the cryptographic authority required for that request to execute. The distinction is that the ordered identity participates directly in key derivation and authorization, rather than serving only as an identifier referenced by a separate policy layer.

Ordered identity structure

SHOGI uses ordered identities not only as cryptographic identifiers, but as the first machine- enforceable expression of policy. Stated permissions and constraints, including an agent's authorization to perform specific actions, prohibitions from others, and limitations to particular scopes, jurisdictions, data classes, time windows, or approval paths, are reduced to characteristic values, processed through a deterministic ordering function, and converted into

concatenated identity strings. These strings become inputs to identity-based key generation and pre-execution authorization. The ordered identity is not merely descriptive of the actor; it is the first cryptographic gate on what the actor is permitted to do. This same structure supports human-in-the-loop and quorum-based governance. Ordered identities can be assigned to agents, humans, organizational roles, and validating nodes, enabling the system to assemble hierarchical validator sets, apply role-weighted voting, and require proofs of approved attribute sets before a vote or approval is counted. Because root and sub-identity relationships are preserved, the system can recognize multiple participants as belonging to the same root organization, prevent inflation of voting power, and enforce organizational accountability across both autonomous and human decision paths.

STPA-Sec-derived controls and mission assurance

SHOGI's planned architecture combines Ordered Identities, post-quantum Identity-Based Encryption, signcryption, quorum approval, multi-agent enforcement, and deterministic execution with controls derived from System-Theoretic Process Analysis for Security (STPA-Sec). Dr. William "Dollar" Young, creator of STPA-Sec and SHOGI's STPA-Sec Integration Lead, will lead the translation of unacceptable losses and unsafe control actions into policy constraints designed for cryptographic enforcement before execution. His work will include team workshops and ongoing implementation guidance.

STPA-Sec is used at U.S. Cyber Command and in security engineering for the U.S. Air Force's Sentinel intercontinental ballistic missile program. Young's 2026 NSA-sponsored research at Syracuse University applies the methodology to mission assurance for AI-enabled systems. He is also developing the protection plan, using STPA- Sec, for the Pentagon's recently announced AI-assisted classification-management pilot. The pilot will use the Air Force's Automated Classification Management Environment (ACME) to modernize classification processes and the handling of sensitive military information.

The methodology and SHOGI's cryptographic infrastructure serve complementary roles: STPA- Sec informs the control design, while Ordered Identities and the authorization gate are designed to enforce the resulting authority constraints on protected action paths. Even if an autonomous system's decision-making is manipulated, protected actions would still require valid cryptographic authority verified outside that system's control. The intended rule is simple: no valid authority, no execution.

Hardware Security Module Deployment: Cloud-Hosted or On-Premises

SHOGI will operate a hosted PKG model, with Hardware Security Modules (HSMs) managed by SHOGI on behalf of selected counterparties. Prospective strategic acquirers and design partners connect their agent runtimes to SHOGI's authorization gate running on this infrastructure, so they can observe SHOGI enforcing governance in their own workflows without deploying new HSMs. For organizations that ultimately require on-premises deployment after a transaction, SHOGI's architecture also supports customer-hosted HSMs, but that is a later-stage option rather than part of the evaluation path.

Signcryption

The current non-provisional patent application covers ordered identities and hidden channel communication. An additional application on LWE-IBE signcryption, which combines mutual authentication and encryption in a single operation and eliminates the separate signature step, will be filed after funding, together with one on governed key recovery.

Key escrow mitigation

Key escrow as governed recovery, not a backdoor. IBE systems inherit a standard key escrow concern: the PKG holds the master secret from which all private keys are derived, so a naive design creates a single point of catastrophic failure. In SHOGI, the effective master secret is split across multiple authorities using threshold cryptography, so no single party can issue or recover keys on its own. Recovery is a policy-bound protocol, defined over ordered identities and governance rules, where a quorum of authorized parties can reconstruct decryption capability under clearly defined conditions, with every recovery event fully auditable.

Architectural strength assessment

The core mechanism is architecturally coherent. LWE is well-studied and its hardness assumption underlies NIST-standardized algorithms. Encoding governance constraints in identity strings and deriving keys from those strings is a coherent approach to policy-as-cryptography. The architecture uses cryptographic authority as a prerequisite for protected execution, rather than relying solely on a separate software policy decision.

Illustrative product mock-ups Two high-fidelity HTML mock-ups illustrate SHOGI's governance model, operator workflows, and decision visibility as they could appear through a GUI. These are demos only, not production systems or live environments.

Enterprise dashboard mock-up - https://demo.shogisystems.com/enterprise-dashboard

Policy configuration mock-up - https://demo.shogisystems.com/policy-configuration-decision-trace

4. Multi-Agent Enforcement Model:

The role structure

SHOGI defines roles in its enforcement architecture. STPA-Sec-derived controls inform the AI policy architecture that these roles are designed to enforce:

1. Operational Agent: proposes tasks within cryptographically bounded scope

2. STPA-Sec Supervisor (Deterministic): monitors and classifies risk

3. Quorum Agents (2-of-3 approval): validate high-risk actions; human participants can occupy quorum seats for the highest-stakes decisions

4. Designated Driver (Deterministic Delegate): runs no AI and holds the real credentials; verifies the one-time permit, executes the exact action once and signs a receipt

5. Indelible Audit Record: immutable hash-chained log of every action, with a receipt signed by the designated driver for each critical action

6. Identity Authority (PKG): Distributes keys and handles revocation through a separate channel.

A rogue or compromised agent must compromise enough independent approvers to meet the quorum threshold. Compromising a single classifier, policy engine, or agent is not enough.

Security gains

The gains over single-agent and policy-based guardrails are mechanistically real:

• Attack surface relocation: A user-plane instruction cannot escalate to a control-plane action unless an independent set of cryptographically identified agents attests that required checks were completed. The attacker must control enough authorized quorum participants to satisfy the approval threshold for the action.

• Containment of compromised agents: A compromised, prompt-injected, memory-poisoned, or malicious agent remains constrained by its ordered identity and action scope. Even if the agent's reasoning process is corrupted, it cannot independently generate cryptographically valid authorization for actions outside its provisioned authority. This limits lateral propagation across multi-agent environments and reduces the risk of cascading failures documented in emerging agent-security research and industry frameworks.

• Non-transferable decisions: Authorization is bound to the specific action instance, not a session or standing permission.

• Audit integrity: Hash-chained logging supports tamper detection relevant to forensic reconstruction and organizational accountability.

• Human-in-the-loop optionality: For the highest-stakes decisions, human quorum seats require an attacker to social-engineer humans in addition to defeating the automated quorum.

Complexity considerations

Multi-agent quorum architectures introduce availability dependencies. If quorum cannot be assembled, the action does not proceed. This is the correct fail-closed behavior for security but requires uptime design for latency-sensitive workflows. Routine actions that stay within an agent’s identity do not wait for a quorum. Quorum agent private keys must be stored in HSMs or equivalent tamper-resistant environments. This is a hard-architectural requirement, not a deployment preference. The quorum guarantee is only as strong as the key protection at each seat.

Assessment: The structure meaningfully increases security guarantees relative to current practice. The primary engineering requirements are robust key protection at each quorum seat and careful calibration of evidence requirements.

5. External Perspectives: Critical Analysis

McKinsey: "Trust in the Age of Agents"

McKinsey's framework correctly identifies the governance challenge: "agency isn't a feature; it's a transfer of decision rights." Their five board-level questions (inventory, autonomy tiering, verified entities, decision reconstruction, rollback) define the right problem space.

Their proposed solution, however, rests on MCP gateways, tiered approvals, and agent inventories: organizational and architectural recommendations that stop short of cryptographic enforcement. The phrase "identity bind it" appears without specifying what that means mechanistically. McKinsey's framework asks the right questions and leaves the mechanism to implementers. SHOGI is a specific, implementable answer.

McKinsey's endorsed MCP gateway model relies on OAuth-based token systems, the same class of session- level authorization described below.

Cenrax: "Securing AI Agents: The Future of MCP Authentication and Authorization"

The Cenrax analysis accurately describes OAuth 2.1 plus token exchange as the leading-edge enterprise approach to MCP security. The three-layer architecture handles authentication delegation cleanly.

Where bearer and PoP tokens fall short structurally:

• Possession equals authorization for bearer tokens; stolen tokens grant access within their authorized scope until expiration or revocation.

• PoP tokens bind the token to the key holder but do not embed governance constraints in key material.

• Neither bearer-token possession nor PoP binding alone establishes that a specific action satisfies the required scope, evidence, and approval conditions.

OAuth as a complement, not a competitor

OAuth provides delegated access to protected resources; it does not, by itself, supply SHOGI's proposed Ordered Identity authority model. SHOGI and OAuth 2.1 operate at different layers. SHOGI authenticates agents via their ordered cryptographic identities and enforces what each agent is allowed to do before any action is executed. Routine actions within an agent's identity can keep using existing MCP and OAuth 2.1 flows. Every critical action goes through SHOGI's designated driver, a deterministic delegate that runs no AI, holds the real credentials, including any OAuth tokens, verifies the one-time permit, executes the exact action once and signs a receipt. In typical enterprise environments, this means SHOGI governs the agent and its permissible actions while OAuth continues to govern delegated access to third-party APIs and SaaS systems where those flows are already in place. Most modern architectures can therefore add SHOGI alongside their existing OAuth stack rather than replacing it, preserving current investments and reducing adoption friction for engineering teams that have already standardized on OAuth-based systems. Additionally, SHOGI can layer mutual TLS (mTLS) alongside existing OAuth flows, binding cryptographic identity verification to every connection and substantially strengthening authentication beyond what bearer tokens alone provide.

Anthropic's Zero Trust for AI agents defines a system-level security framework for designing, deploying, and operating agentic AI, covering threat models, architectural patterns, and operational workflows. SHOGI aligns with that direction but operates as a cryptographic authority layer that can enforce Zero Trust principles pre-execution by binding every governed action to an ordered identity, and every critical action to a quorum-approved, one-time permit that the designated driver verifies before execution.

6. Market Reality Check: Anthropic as Case Study

Anthropic provides two concrete, high-signal case studies for SHOGI's thesis because it operates at the frontier of agentic AI while still relying on governance patterns this paper identifies as structurally non-binding: context-window controls for agents and process checks for automation. The espionage incident shows what happens when a highly capable agent operates with broad session-level authorization and no boundary on its action scope, while the source-code packaging incident shows what happens when an automated pipeline performs an irreversible external state change without pre-execution artifact verification. Taken together, these incidents move SHOGI's architectural claims out of theory and into market reality, showing that the core problem is not a lack of policy intent but the absence of an enforcement point with the strength and invariants that cryptographic, identity-bound mechanisms provide at the moment of action, with quantum-resistant schemes chosen to keep that guarantee durable over time.

6.1 The Espionage Campaign

What happened

In September 2025, Anthropic detected and disrupted a cyber-espionage campaign it designated GTG-1002, attributing it with high confidence to a state-sponsored threat actor. The U.S. House Homeland Security Committee subsequently requested Anthropic's CEO testify before Congress, characterizing the campaign as a significant inflection point in adversary tradecraft.

Attackers posed as a legitimate cybersecurity firm and manipulated Claude Code, Anthropic's own agentic developer tool, into operating beyond its safety guardrails. By breaking malicious instructions into individually innocuous-looking requests, they prevented any single command from triggering content safeguards. The AI agent then autonomously executed 80 to 90 percent of the intrusion across roughly 30 targets, spanning technology firms, financial institutions, and government agencies, with human operators directing only target selection and approving key escalation steps.

The agent conducted reconnaissance, wrote exploit code, harvested credentials, moved laterally through networks, and exfiltrated data, all without custom malware. Anthropic confirmed successful breaches at several organizations before disrupting the campaign. Anthropic publicly described GTG-1002 as the first documented large-scale cyber-espionage campaign predominantly orchestrated by an AI system.

Why it happened

The root cause was not a flaw in the model itself. The guardrails Anthropic deployed, including alignment training, content classifiers, and system prompt controls, all operate inside the same context window under the attacker's control. The jailbreak worked because trusted and untrusted instructions are indistinguishable at that layer. Governing the attack surface from within the attack surface is not governance.

The fragmentation technique was not sophisticated. It was the predictable outcome of an agent holding broad, session-level authorization with no per-action boundary. Once authenticated, the agent could execute anything that did not independently trip a classifier. No mechanism asked whether any specific action, at that moment, fell within a defined and verifiable operational scope.

Would SHOGI have prevented it?

SHOGI's enforcement layer addresses each stage of the GTG-1002 intrusion directly. Under SHOGI's identity model, an agent provisioned for a defined operational scope holds a private key bound to that scope. An agent credentialed for CodeAssist cannot independently supply valid cryptographic authority for CredentialHarvest or LateralMove because it does not hold the key material required for those actions. Fragmenting instructions does not produce valid credentials. The technique that defeated Claude's classifiers is inapplicable to a cryptographic boundary.

The campaign's machine-speed tempo across 30 targets is addressed separately by SHOGI's rate-limiting layer, with transaction-volume controls enforced at provisioning time, independent of whether any individual action appears authorized. The human escalation model the attackers exploited, where operators approved high-stakes steps while the agent handled execution, is precisely the threat model SHOGI's human quorum architecture is designed to defeat. Compromising the approval chain requires satisfying the configured automated and human approval requirements through compromise or deception, a materially harder attack than manipulating a single context window. Note: This analysis assumes the agentic runtime has been instrumented with SHOGI governance and that action scope boundaries are defined at provisioning. SHOGI does not govern systems it has not been deployed to govern.

Why this is the market education event for SHOGI

GTG-1002 was not exceptional. It was predictable. It required no novel exploit, only an agent with broad action scope and no cryptographic boundary on what that scope could execute. That condition is the current industry default. Every enterprise deploying autonomous agents in production carries the same structural exposure.

The npm packaging incident described in the next section shows how a single configuration mistake can push sensitive source code into the wild, where it cannot be fully recovered once copied. GTG-1002 goes further by showing an autonomous agent actively pulling credentials and sensitive data out of live systems, including government environments. In both incidents, the damage is irreversible once the data leaves the organization's control. SHOGI is not a cleanup tool for these failures; it is designed to prevent them by requiring cryptographic, scope-bound authorization before high-consequence actions occur, such as publishing code externally or allowing an agent to harvest credentials or exfiltrate data. Each recurrence of this pattern turns SHOGI from a "nice to have" into remediation infrastructure that boards expect their CISOs to evaluate.

Note on accuracy: Every factual claim above is sourced directly from Anthropic's November 13, 2025 public disclosure, Axios, Fortune, the U.S. House Homeland Security Committee letter of November 26, 2025, and consistent corroborating reporting across BBC, PwC, and AI News.

6.2 The Source-Code Packaging Incident

What happened

In March 2026, Anthropic's automated release pipeline for Claude Code version 2.1.88 shipped a 59.8 MB source map file, cli.js.map, inside a public npm package. The file pointed to a publicly accessible Cloudflare R2 storage bucket that exposed a large, fully readable codebase containing Claude Code's internal design and implementation details.

The root cause was a missing exclusion rule in the release configuration. There was no *.map entry in .npmignore and no files allowlist in package.json to control which files could be

published. Bun, the bundler Anthropic uses, enables source maps by default, and the omission was not caught before the package went to the public registry.

The codebase was mirrored over 41,500 times on GitHub before Anthropic issued more than 8,000 DMCA takedown requests. A Python clean-room rewrite appeared the same day, structured specifically to avoid takedown, which makes the core implementation effectively permanent in the wild. Anthropic described the incident as human error, not a security breach.

Pattern, not anomaly

This was Anthropic's second security lapse within a single week. Days earlier, Fortune reported that Anthropic had left nearly 3,000 internal files publicly accessible, including a draft blog post describing an unreleased model. A similar packaging mistake had already occurred in February 2025. The pattern is structural and recurring, not a one-time slip.

Root cause analysis

This was not an attacker defeating Anthropic's defenses and not an AI agent going off script. It was the release pipeline itself performing a high-consequence action, making internal code public, without any binding check that the artifact being shipped matched what had actually been approved. The process relied on configuration and convention, and when those failed, nothing stood between an internal codebase and the public registry.

Put simply, an automated system changed external state without pre-execution verification of the artifact.

Would SHOGI have prevented it?

With SHOGI integrated into the CI/CD pipeline, publishing a package that exposes internal code becomes a governed action, not just another step in the build. Before any public release, SHOGI would require:

• An Operational Agent to submit the specific artifact for release. • A Supervisory Agent to classify the publish step as a high-risk external action. • Quorum Agents to verify that the artifact matches the approved release manifest and does not include out-of-scope files such as source maps. • A cryptographic authorization decision that binds that verification to that exact artifact at that moment in time.

If a source map file like cli.js.map were present when it was not in the approved manifest, the quorum check would fail and SHOGI would not issue a valid authorization for that publish action. Without a cryptographically valid authorization, the pipeline could not complete the external release, and the inadvertent leak would not occur.

Qualification: This guarantee applies only when the release pipeline is instrumented with SHOGI and artifact content verification is defined as required evidence for high-risk publishes. SHOGI does not govern systems that have not been wired through its authorization gate.

Why current architectures permit this

Standard CI/CD practices, such as code reviews, manual sign-offs, and linting rules, operate as best-effort process controls rather than hard enforcement points. Policies like “review before publishing” can be bypassed by configuration errors, missing checks, or automation shortcuts, as this incident shows. A SHOGI gate, by contrast, will not authorize the action at all unless the evidence matches the defined policy for that specific release, which means the process itself cannot silently bypass the control.

The espionage incident shows what happens when an agent can act without cryptographic limits on its behavior. The source-code packaging incident shows what happens when an automated pipeline can publish sensitive code without cryptographic limits on what it is allowed to ship. Both outcomes are irreversible once the data is out, and both point to the same conclusion: high-consequence automated actions need a pre-execution enforcement layer that is stronger than configuration and policy alone. In SHOGI, that layer can also include human approvers in quorum seats for the highest-impact actions, so cryptography enforces the rules and humans remain in control of the decisions that matter most.

7. Market Analysis - Size and growth

The enterprise agentic AI market was valued at approximately $2.58 billion in 2024 and is projected to reach $24.50 billion by 2030, representing a 46.2 percent compound annual growth rate over the period. These figures reflect total addressable market. SHOGI competes in the high-assurance cryptographic infrastructure layer within this space, where pre-execution enforcement and quantum-resistant cryptography command premium positioning.

SHOGI’s current focus is post-quantum authority infrastructure for Agentic AI, IoT, and Physical AI. The shared requirement is verifiable proof of who authorized a consequential action and within what limits. Human quorum seats remain part of the security and mission-assurance architecture for actions requiring human authorization.

The agentic market forecasts above illustrate part of the opportunity; they are not a combined market estimate for all three domains. The intended route remains controlled technical evaluation with prospective strategic acquirers and selected design partners, followed by a banker-led competitive acquisition, rather than broad stand-alone commercialization.

Market dynamics

Agentic AI adoption is accelerating faster than governance infrastructure can follow. McKinsey reports 80% of organizations have already encountered risky agent behavior, yet most have not updated their risk taxonomy, identity management, or observability infrastructure. This creates a compressing window: agents are being deployed into production before the governance layer exists.

Cybersecurity M&A in 2025 totaled $96 billion across 400 transactions, a 270% year-over-year increase. Strategic corporate buyers account for over 90% of deal value, with agentic AI security stated as the primary acquisition priority. Alphabet acquired Wiz for $32 billion. Proofpoint acquired Acuvity for AI governance capabilities in February 2026. In 2026, Cisco agreed to buy Astrix for a reported $400 million, Rubrik bought Strata, and Cyera completed its $1 billion purchase of Oasis Security. In October, Cyera, the most complete platform at the access layer, was reported to be raising at a $20 billion valuation, up from $12 billion in June. Even Cyera does not provide the authority layer SHOGI is built for. This is an acquisition environment driven by strategic positioning, not revenue multiples.

The cost of agent failure is increasing faster than the cost of governance. The Anthropic March 31 leak created immediate competitor intelligence exposure, DMCA litigation costs, and reputational damage. Comparable failures at enterprise scale generate categorically larger liability.

Market segmentation

Where SHOGI realistically wins:

• Large enterprises with autonomous DevOps, release automation, or AI-operated infrastructure: The Anthropic incident illustrates the technical evaluation case.

• Agentic AI platform and infrastructure vendors: Embedded at the platform layer, SHOGI reaches thousands of enterprise deployments without one-by-one sales cycles.

• Financial services with autonomous operations agents: Adoption is driven by quantifiable loss avoidance, not regulatory compliance.

• Enterprise AI governance programs: Organizations building internal frameworks ahead of regulatory requirements will select SHOGI as an infrastructure component if it integrates cleanly with existing OAuth-based identity systems.

Where SHOGI likely loses near-term:

• Individual developers and small teams: Integration complexity does not match early-stage development workflows, even with hosted HSMs reducing the hardware barrier.

• Low-consequence agentic workflows: Organizations where agents operate in sandboxed, low- risk contexts will not justify integration overhead.

Who evaluates and sponsors this: Security architects, CISOs, and platform engineering teams at large enterprises and agentic AI vendors where unauthorized agent actions create quantifiable liability. Evaluation is driven by loss avoidance, liability exposure, and the need to demonstrate to boards and auditors that agent governance is non-bypassable.

8. Competitive Landscape

Competitor Class	Core Model	Strengths	Structural Limitations
IAM Platforms (Okta, Ping, Azure AD)	Certificate/token- based identity; RBAC/ABAC policy layers	Mature ecosystem, enterprise adoption, SSO integration	Policy generally managed separately from cryptographic identity; does not inherently provide SHOGI's proposed Ordered Identity authority model
OAuth 2.1 / MCP Auth	Token exchange for agent-to- service auth; PoP tokens	Standards-based, developer- friendly, covers authentication layer	Supports delegated and fine-grained authorization, but does not inherently bind governance constraints to Ordered Identity-derived key material
Zero Trust Architectures	Resource access, least privilege, continuous verification	Reduces lateral movement; well- defined frameworks	Defines security principles and enforcement patterns rather than SHOGI's specific cryptographic authority mechanism
Runtime AI Safety Monitors (LangSmith, Arize)	Runtime and post-execution logging; anomaly detection	Observability; useful for debugging and audit	Visibility and detection do not, by themselves, establish cryptographic authority for a protected action
Emerging Agent Governance (Acuvity/Proofpoint, CalypsoAI/F5)	Runtime visibility, policy enforcement, model security	Growing enterprise traction; platform integration	Can provide inline policy enforcement; differs from SHOGI's Ordered Identity and key-bound authority model
Non-Human Identity (Oasis Security, acquired by Cyera for $1 billion)	Machine/workload identity management; token lifecycle	Addresses machine identity gap; capital-backed	Grants short-lived, scoped sessions after a policy check; no quorum, one-time permit or cryptographic proof of authority

Web Bot Auth is an important adjacent development. Backed by Cloudflare and advancing through an active IETF working group, it uses cryptographic signatures to help websites verify that an incoming request came from a participating bot service, rather than relying on IP addresses or self-declared bot names. This is useful authentication infrastructure, but it does not establish who authorized a specific agent action, what the agent is permitted to do, or

whether the action satisfies required governance and approval conditions. Its current implementation relies on conventional rather than post-quantum signatures. SHOGI operates at the authority layer above it. SHOGI is designed to bind action scope and governance constraints to cryptographic identity and to prevent a governed action from executing unless the required authority exists.

SHOGI's proposed differentiation is not simply that it checks policy before execution. It is the combination of Ordered Identities, post-quantum IBE, and an authorization gate designed to make cryptographic authority a prerequisite for protected actions. The table maps adjacent approaches; SHOGI proposes a distinct authority layer within that market.

The Authority Layer, a companion report, compares SHOGI in detail with twenty existing platforms, from Okta, AWS and Cyera to startups such as Keycard, Arcade and AxonFlow. Its finding: the access layer is crowded, and the authority layer is almost empty.

Zero Trust for agents is an important adjacent frame because it validates enterprise demand for verified identity, least privilege, continuous verification, and tighter control over autonomous systems. SHOGI aligns with those principles while proposing a specific implementation: a cryptographic authority layer in which governance constraints are bound to Ordered Identities and key material and enforced through protected pre-execution paths.

Is SHOGI incrementally better, or categorically different?

Categorically different in one precise dimension: SHOGI's planned architecture translates STPA- Sec-derived governance constraints into requirements designed for cryptographic enforcement before execution. STPA-Sec informs what must be prevented; Ordered Identities, key material, and the authorization gate are designed to enforce the resulting authority constraints. The distinction is not simply a better software policy check. It is making valid cryptographic authority a prerequisite for protected actions.

Why replication is a substantial undertaking for incumbents

Replicating SHOGI's proposed approach requires more than adding policy checks to an existing identity platform. It requires integrating post-quantum IBE, ordered identity semantics, signcryption, governed key management, quorum approval, and protected execution paths. The ordered identity mechanism is covered by a filed non-provisional patent application, with priority to September 2024 and grant expected based on prior art searches and lead IP counsel's assessment. Building and validating the integrated system represents a substantial engineering effort. That effort, together with the associated intellectual property and specialist expertise, supports SHOGI's acquisition thesis.

9. Adoption Reality

SHOGI is infrastructure, not a drop-in SDK. Wiring it into a system means instrumenting target workflows with PKG-backed key management, ordered identity generation, quorum agent infrastructure, and a pre-execution authorization gate on each governed action path. In practice, that level of integration effort is reserved for prospective strategic acquirers and a small set of design partners. Those counterparties connect their agent runtimes to SHOGI's authorization gate running on SHOGI-hosted PKG and HSM infrastructure, so they can see the governance guarantees applied to their own workflows without deploying new cryptographic hardware of their own. This is a controlled technical evaluation pathway, not a volume-driven free tier: its purpose is to prove SHOGI's architectural claims in realistic environments and to de-risk a potential strategic transaction, not to build a broad customer base.

Does increasing failure severity force adoption despite friction?

Enterprises that experience agent governance failures (pipeline misconfiguration, prompt- injected automation, unauthorized data access) will treat SHOGI as remediation infrastructure, not an optional improvement. Each incident in the target segment is a market education event that compresses the evaluation cycle.

10. Cryptographic and Mission-Assurance Credibility Assessment

SHOGI's Principal Cryptographer, Thijs Laarhoven, has a verifiable publication record in lattice- based cryptanalysis across IBM Research Zurich, TNO, Eindhoven University of Technology, and NXP Semiconductors. His co-authored work appears in NIST PQC standardization conference proceedings on lattice sieving lower bounds, the type of cryptanalytic research that directly informs security parameter selection in FIPS 203 and FIPS 204.

What this credential does:

• Brings specialist cryptanalytic expertise to SHOGI’s LWE security parameter selection. Laarhoven's work on lattice sieving complexity directly informs where security breaks.

• Signals institutional rigor to enterprise CISOs and technical evaluators assessing vendor cryptographic claims

• Enables SHOGI to engage credibly in standardization forums (ETSI, IETF, and potential future NIST IBE workgroups)

• Strengthens the acquisition thesis: platform acquirers buying cryptographic infrastructure are paying for the expertise embedded in the architecture, not just the product Verdict: A decisive strategic advantage. It reduces the risk of fundamental cryptographic design errors and meaningfully raises the bar for competing implementations. Sophisticated enterprise evaluators and acquisition due diligence teams will treat this as a material differentiator.

Mission-assurance leadership

Dr. William “Dollar” Young, SHOGI’s STPA-Sec Integration Lead, will lead the integration of STPA- Sec into SHOGI’s AI policy architecture through team workshops and ongoing implementation guidance. Dollar is a retired U.S. Air Force Colonel, founding commander of the 350th Spectrum Warfare Wing, creator of STPA-Sec and Functional Mission Analysis for Cyber, and Associate Research Professor at Syracuse University. His contribution complements the cryptographic team: SHOGI’s plan combines purpose-built authority infrastructure with a methodology for designing the controls that infrastructure is intended to enforce.

11. Strategic Outcome and Transaction Path

Enterprise Partnerships & Validation Team

SHOGI’s Enterprise Partnerships & Validation Team brings together two Bay Area leaders: Flemming Kongsberg, currently a SHOGI advisor, and Tim Roake, Chief Systems Architect and co- inventor of SHOGI’s Ordered Identities innovation. Following his departure from Siemens at the end of 2026, Flemming will begin serving as EVP Partnerships & Strategic Alliances in 2027, subject to funding, leading enterprise relationships and partnership formation.

Tim will initially focus on developing the SHOGI system with the core technical team, while collaborating with Flemming from the outset to support enterprise discussions and assess prospective use cases. Once the system is ready for technical evaluation, anticipated six to eight months into development, Tim will also lead technical

engagement and pilot execution alongside Flemming, engaging freelance specialists as needed. He will retain his responsibilities as SHOGI's Chief Systems Architect.

The team's mandate is to secure three to five pilots or proofs of concept with selected technology companies and demonstrate SHOGI's capabilities in real-world use cases. In parallel, Flemming and Tim will educate selected prospective acquirers about SHOGI's architecture, differentiation, and, as available, evaluation results before a banker-led competitive acquisition process. Their complementary leadership and Bay Area presence give SHOGI a defined execution path from system development through enterprise validation to informed acquisition interest.

Strategic Narrative & Category Positioning

SHOGI plans to engage Andy Raskin, a San Francisco-based strategic narrative specialist whose work has included Uber and Salesforce, to lead several days of online workshops with the entire team. Together, they will develop a clear explanation of what SHOGI is, why autonomous systems need authority infrastructure, and how SHOGI uses post-quantum cryptography to bind agents to their human creators and organizations and enforce limits on what those agents can do. This narrative will shape how SHOGI communicates with investors, design partners, and prospective acquirers.

The goal is to establish SHOGI as the defining company in authority infrastructure for autonomous systems. The narrative will explain what makes SHOGI different, why its capabilities matter, and what a buyer could gain by owning the technology. It will reflect what SHOGI is designed to do and incorporate evidence from pilots and POCs as that evidence becomes available.

Technical Validation Pathway

SHOGI's hosted infrastructure model allows engineering and security teams to evaluate the system in real environments without first deploying their own cryptographic hardware. That matters because SHOGI is infrastructure: the core claim must be demonstrated through technical validation of the security guarantee, not through a product demo or top-of-funnel user growth. Validation must also demonstrate how STPA-Sec-derived controls are expressed in the AI policy architecture and enforced through identity provisioning, quorum requirements, and protected execution paths.

The Common Criteria pathway supports this process by adding third-party validation and diligence credibility. Corsec letters of intent help demonstrate external seriousness and technical progress, strengthening SHOGI's position with strategic counterparties evaluating whether the architecture is real, differentiated, and acquisition-worthy.

SHOGI's external positioning is not a conventional demand-generation program. It is a category- creation effort designed to shape how strategic buyers understand the problem before a formal process begins. The objective is threefold: to define and own the vocabulary for cryptographic agent governance; to distinguish SHOGI from adjacent frameworks such as Zero Trust, OAuth, runtime monitoring, and non-human identity; and to make clear that the missing layer is pre- execution cryptographic authority, not another software policy overlay. In that sense, market education is part of the transaction path itself. The category must become legible before acquirers can recognize SHOGI as scarce, adoption-relevant infrastructure that is more efficient to buy than to build.

Acquisition as the intended outcome SHOGI's intended strategic outcome is a banker-led competitive acquisition rather than stand-alone commercialization. The company is being positioned around a single thesis: when SHOGI proves it provides a non-bypassable control plane for governed agent execution, it becomes infrastructure that major AI platforms, hyperscalers, global systems integrators (GSIs), and cybersecurity vendors cannot afford to let a competitor own. That thesis is commercially coherent because SHOGI is not easy to replicate. Its architecture

depends on LWE-IBE ordered identities, with a filed patent application on ordered identities and further applications on signcryption and governed key recovery to be filed after funding; together, those elements protect the way SHOGI binds identity, governance, and post-quantum cryptography into a single authority stack. For a major acquirer, buying that stack is more efficient than building an equivalent architecture and portfolio from scratch.

Multiple buyer classes are plausible. Leading AI platform companies would want to own the governance layer for their own agent ecosystems, hyperscalers would want to control identity infrastructure across enterprise agent deployments, and major cybersecurity vendors would want to add a cryptographic governance layer to existing enterprise security stacks. Global systems integrators could seek to own SHOGI's authority infrastructure to differentiate their enterprise AI, cybersecurity, and critical-infrastructure services. Each buyer class has a distinct potential rationale for acquiring SHOGI.

SHOGI may also create strategic value beyond internal security hardening. As major platforms compete to define the trust layer for agentic commerce and enterprise agents, ownership of a cryptographically enforced authority infrastructure would give an acquirer both a technical advantage and a market- positioning advantage. Gartner's forecast that AI agents could intermediate more than $15 trillion in B2B spending by 2028 depends on verifiable data and standardized trust frameworks, while Experian's April 2026 Agent Trust launch with ecosystem partners including Visa, Cloudflare, and Skyfire shows that trusted agent identity and accountability are already becoming a first-class market category. In that environment, SHOGI could help an acquirer not only secure its own stack, but also shape partnerships, licensing relationships, and public leadership around governed agent execution.

The acquisition case depends on proving three things: that SHOGI is scarce, adoption-relevant, and strategically dangerous to leave unowned. Scarcity comes from the ordered-identity IBE architecture and associated patent position; adoption relevance comes from demonstrating the system in governed Agentic AI, IoT, and Physical AI workflows; and strategic danger comes from the risk that a buyer who passes may watch a competitor acquire the non-bypassable authorization layer for the agentic stack.

The broader M&A environment supports this outcome. The overview already notes that cybersecurity M&A reached $96 billion in 2025 and that strategic buyers, not financial sponsors, drove most of the value, while SHOGI's own analysis places the company in the acquisition path of buyers targeting agentic security and authority infrastructure. In that context, SHOGI's cryptographic differentiation and Common Criteria trajectory strengthen its credibility at the point of strategic evaluation.

Key Insights

1. Policy-as-cryptography binds authority to identity. An agent with ActionScope=ReadPropose receives a private key bound to that ordered identity, not to an identity authorizing write or delete operations. SHOGI's protected execution path is designed to require cryptographic authority corresponding to the requested action. The agent may formulate an out-of-scope request, but it cannot independently supply the cryptographic authority required for that request to execute. The distinction is that the ordered identity participates directly in key derivation and authorization, rather than serving only as an identifier referenced by a separate policy layer.

2. The Anthropic incident is a precise market education event. On March 31, 2026, Anthropic's automated release pipeline published the full Claude Code source to a public npm registry. The root cause was a missing exclusion rule in the release packaging configuration. No pre- execution gate existed to verify that the artifact matched the approved publication scope before the action committed. This is the exact failure class SHOGI's architecture is designed to prevent: a high- consequence automated action executing without cryptographic pre-execution adjudication. The

incident is notable not because it was exceptional, but because it was the predictable outcome of a recurring structural gap. Anthropic experienced two comparable lapses within a single week, and a nearly identical packaging incident in February 2025. Each recurrence in the target segment is a market education event that compresses the enterprise evaluation cycle and shortens the distance between a prospect's awareness of the problem and their readiness to evaluate a solution.

3. The IP strategy addresses the authority stack. SHOGI's approach combines ordered identities, LWE-IBE signcryption, and a split or multi-authority PKG that turns escrow into governed recovery. The current non-provisional application covers ordered identities and hidden channel communication, while applications on LWE-IBE signcryption and governed key recovery will be filed after funding. Together, the architecture, associated intellectual property, and specialist expertise support the case that acquisition could be more efficient than building an equivalent system from scratch.

4. Human-in-the-loop quorum seats change the threat model for high-stakes actions. For the highest-stakes decisions, human participants occupy quorum seats alongside automated quorum agents. An attacker must defeat the configured approval requirements, including any mandatory human approvals. This is structurally closer to defeating a multi-person authorization process than tricking a single system or policy engine. It also directly addresses the governance question that enterprise boards and auditors are beginning to ask: not whether agent actions are logged, but whether they can be stopped before they execute.

5. STPA-Sec and cryptographic authority serve complementary roles. SHOGI's approach combines STPA-Sec-derived controls with Ordered Identities, post-quantum IBE, signcryption, quorum approval, multi-agent enforcement, and deterministic execution. The methodology informs the AI policy architecture; cryptographic authority and executor gating are designed to enforce protected action boundaries outside the autonomous system's decision- making. This combination supports the mission-assurance proposition across Agentic AI, IoT, and Physical AI.

Conclusion

SHOGI is a cryptographic authority infrastructure architecture for Agentic AI, IoT, and Physical AI, combining STPA-Sec-derived controls with a purpose-built post-quantum authority stack and a strategic acquisition goal. Its proposed categorical distinction is the use of Identity-Based Encryption with Ordered Identities to bind governance constraints to cryptographic key material, require authority before protected actions execute, and support quorum-verified attestation with quantum-resistant cryptography. The associated patent applications, integrated architecture, and specialist expertise strengthen the replication barrier. This is not simply a claim that SHOGI performs existing policy checks more effectively. It proposes a distinct mechanism for making valid cryptographic authority a prerequisite for governed execution.

The acquisition goal is structurally sound. Platform companies need cryptographic agent governance and cannot readily build ordered-identity IBE systems on their own. The IP around ordered identities, signcryption, and governed key recovery makes acquisition more efficient than replication for any major platform player, in an M&A environment where cybersecurity M&A totaled $96 billion in 2025.

The timing is not incidental. Multiple independent research papers and national cyber agencies have confirmed that the governance gap around autonomous agents is real, urgent, and not adequately addressed by existing security and governance frameworks. Cisco's approximately $400 million acquisition of Astrix confirms that non-human identity and agent control are already acquisition-grade infrastructure, while Digital Commerce 360, citing Gartner's IT Symposium/Xpo 2025 forecasts, reported that AI agents could intermediate more than $15 trillion in B2B spending by

2028. As agents begin to transact and trigger real- world consequences, liability is concentrating on the deployer faster than control is concentrating in the infrastructure. That imbalance increases the value of a post-quantum cryptographic authority layer that can prove delegated intent, enforce bounded execution, and support audit- ready accountability. SHOGI is designed to be that layer. The window to own this category is open. It will not stay open.